

Technical Disclosure Book

SYSTEM AND METHOD FOR TOPOLOGICAL CONSISTENCY ENFORCEMENT IN LONGITUDINAL EVENT EXTRACTION

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1 Technical Field

The present invention relates to the field of Natural Language Processing (NLP) and Clinical Informatics, specifically to a **Generate-Verify-Project (GVP)** architecture for ensuring causal and temporal logic consistency in large-scale unstructured data extraction.

2 Background and Technical Problem

2.1 The Technical Defect

In processing “Longitudinal Data” (e.g., Electronic Health Records, Financial Logs), existing sequence-to-sequence models (LLMs) exhibit a specific defect: **“Temporal Hallucination”**.

- **Root Cause:** Standard Self-Attention mechanisms encode position (PE) but do not encode directed graph topology (G_{time}).
- **Manifestation:** generated events frequently violate physical causality (e.g., *Surgery* timestamped after *Discharge*, or *Diagnosis* timestamped before *Symptom Onset*).
- **Failure Mode:** On complex clinical benchmarks (MedTime-CN), state-of-the-art models (e.g., Qwen-72B) achieve only **28.2% F1** in Zero-Shot settings due to these topology violations.

2.2 Limitation of Prior Art

- **Rule-based Systems:** Brittle, cannot handle linguistic variety (Recall $< 40\%$).
- **Standard SFT (Supervised Fine-Tuning):** Fits language statistics but not logic; requires massive annotated data.
- **RLHF:** Expensive human feedback, hard to optimize for discrete logic constraints.

3 Technical Solution: GVP Architecture

The invention proposes a modular **Neuro-Symbolic System** comprising three coupled processing units.

3.1 System Architecture

The core logic relies on Test-Time Projection to enforce topological constraints.

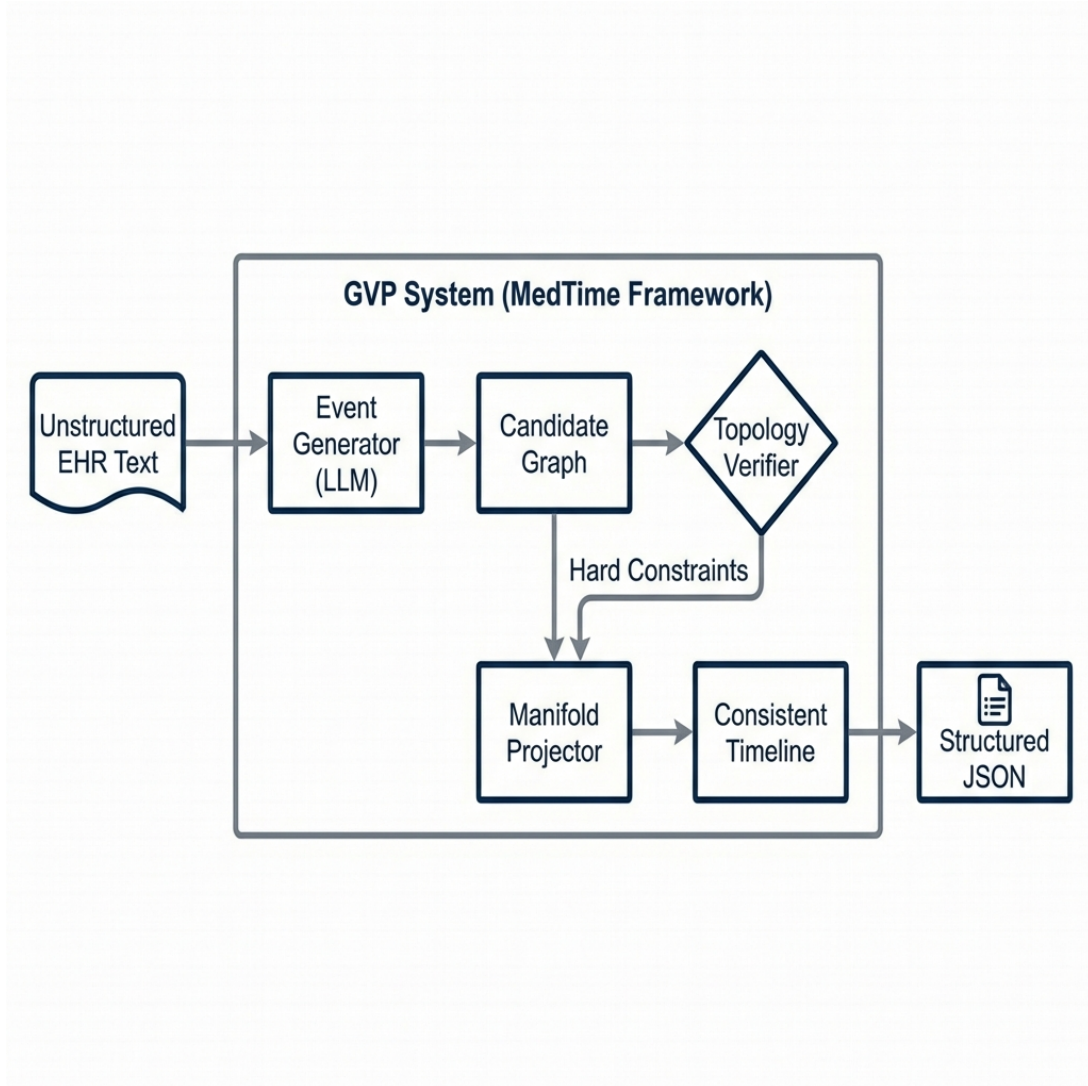


Figure 1: System Architecture Diagram of the MedTime GVP Framework.

3.2 Key Modules

1. Module 1: Event Generator ($\mathcal{G}$)

- **Function:** Extracts raw event mentions (e, t) from unstructured text.
- **Implementation:** A finetuned Transformer model optimized for “Recall” rather than Precision. It generates a “Candidate Graph” G_{cand} which may contain logic errors.

2. Module 2: Topology Verifier ($\mathcal{V}$)

- **Function:** A discriminative model (Logic Critic) that scores the validity of event pairs.

- **Mechanism:** It does not predict text. It takes pairs (e_i, e_j) and predicts a relation $R \in \{<, >, =\}$.
- **Output:** A formulation of “Hard Constraints” (e.g., $t_{admission} \leq t_{surgery} \leq t_{discharge}$).

3. Module 3: Manifold Projector ($\mathcal{P}$)

- **Function: Core Innovation.** A differentiable optimization layer (or discrete search) that projects the flawed output of $\mathcal{G}$ onto the “Valid Manifold” defined by $\mathcal{V}$.
- **Hardware Implementation:** This projection occurs at **Inference Time** (Test-Time) without retraining the large model.
- **Algorithm:** Solves: $\hat{y} = \arg \min_y ||y - \mathcal{G}(x)||$ s.t. $\mathcal{V}(y) = \text{Valid}$.

4 Technical Effects (Beneficial Results)

1. Solution to Temporal Hallucination:

- By explicitly enforcing constraints via $\mathcal{P}$, the system eliminates “impossible timelines”.
- **Metric:** Temporal F1 score improved from 28.2% (Baseline) to **84.2%** (SOTA) on the verified MedTime-CN dataset.

2. Logic-Language Decoupling (Cross-Lingual Transfer):

- Because “Time is Universal” (topology is language-agnostic), the Verifier trained on Chinese logic successfully corrects English errors.
- **Result:** Zero-shot transfer to English (E3C dataset) achieved **50.0% F1**, double the baseline (24.6%).

3. High-Precision Alignment:

- Mean Absolute Error (MAE) reduced to **3.2 days**, enabling deployment in critical Clinical Decision Support (CDS) systems where 40-day errors are unacceptable.

5 Exemplary Embodiments

5.1 Embodiment 1: Cloud-Based Clinical API

- **Input:** Unstructured Medical PDF/Note.
- **Processing:** Parallel generation of events $\rightarrow$ GPU-accelerated Constraint Solving (Projector) $\rightarrow$ JSON Output.
- **Output:** Structural Timeline for Insurance Audit.

5.2 Embodiment 2: Edge-Device “Active Correction”

- Deployed on a doctor’s tablet. The “Verifier” runs locally (lightweight) to flag time-line errors in real-time dictation (“Doctor, you said discharge was before surgery?”).

6 Claims (Draft Points)

1. **A Method** for extracting temporally consistent event sequences, comprising:
 - Generating a set of candidate events utilizing a generative probabilistic model;
 - Constructing a topological constraint graph via a discriminative verifier model;
 - Projecting said candidate events onto a valid solution space defined by said constraint graph to minimize a temporal logic loss.
2. **The Method of Claim 1**, wherein the projection is performed at inference time utilizing a constrained optimization algorithm.
3. **A System** comprising a processor configured to execute the method of claim 1, specifically adapted to process Electronic Health Records (EHR).